ADVX

2026

BOARD ① · Architecture

Nervus OS

Robot's Android

Android for Robots — an OS & app store for any machine

Any brand robot, download and install App like a phone —— One set of App runs on different bodies

Nervus OS Team · ADVX 2026 Hackathon · RDK S100 · Ubuntu 22.04 · aarch64 · PREEMPT_RT

github.com/Nervus-OS

DESIGN CONTRACT · DESIGN AGREEMENT

Switch to another robot, only replace one Provider, core zero changes

Swap the robot → swap one Provider.
The kernel never changes.

4	2	3	∞
Layers	Bus ports	Sided market	App slots
Layer Control Stack	Port protocol	Manufacturer • Developer • User	Open access

Runtime Architecture

Runtime Architecture BETWEEN EACH LAYER IS **REAL ONLINE PROTOCOL** · NOT A FUNCTION CALL

L1 · Top Layer

Application Layer

Applications

Anyone can write an App for the robot. Each App runs separately, manages its own permissions, and requires one-time authorization during installation.

Open Source SDK · nervus-app-sdk · .nspkg Signed Installation

Ring-driven robotic arm
Ring → PiPER · flagship demo

Alien • Limo • Collaboration
Alien · Limo · Collaboration

Actions SetsAction Sets

Open Source SDK • Everyone Can Build Robot Apps
open Kotlin SDK
Open up system capabilities—individuals, companies, can all start building their own robot apps.

More Apps Coming Soon
more landing…

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Port A · Port A — nervus-ipc | UDS · protobuf Envelope | Request / Response | SO_PEERCRED Identity | AcquireControl Lease

L2 · Central Layer · System Kernel

Nervus OS

nervud · Go kernel · ~16k lines

The core of the whole board. All App instructions must go through here: bus, permissions, arbitration, package management, and capability routing are all inside the kernel. It only schedules and arbitrates, **does not touch** high-frequency physical control — that's DimOS's job.

nervud · nervus-ipc · nervus-system-server

Control Focus Arbiter
internal/control
Who currently holds the body · Priority SAFETY>HUMAN>AI · Can be preempted, can be audited.

Three-layer Safety
internal/safety
Emergency stop FIFO 95 · halt/settle · Generation number latching, bypassing DimOS directly to execution layer.

Pack Atomic Motion Gate
internal/motiongate
Motion state machine packed into an atomic word + generation number rollback, take over immediately effective.

Permission Model
internal/permission
Interface → Permission Directory · Decided by permission intersection at installation.

IPC Control Plane
internal/ipc
UDS + Protobuf Envelope · SO_PEERCRED/PEERPIDFD Take trusted identity.

Real-time Scheduling
PREEMPT_RT
sched_setattr SCHED_FIFO · Emergency stop channel zero heap allocation.

Package Manager
package manager
.nspkg Ed25519 signature · Install/Uninstall —— Application store backend.

Capability List + Endpoint Resolution
profile · resolve
Declare device and capability · Capability → Provider · Capability → Resource handle.

Model-agnostic AI Gateway
tool-calling
System capability encapsulated as tool, any brain driving any body.

Port B · Port B — Capability → Provider Binding | Robot Profile → Provider | Ability → Drive Instantiation | Manipulator@1 · BaseMotion@1 | Switch robot = Switch Provider · Core remains unchanged

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L3 · Driver Adaptation Layer

DimOS

Dimensional OS

Hardware Manipulation and Autonomous Navigation. Located below Nervus OS **as a Provider**, uniformly under its control.

dimensionalOS/dimos · Pure Python · No ROS

Unitree GO2 Provider
unitree-go2

SLAM • A* Planning
navigation

Manipulator Provider
PiPER · NERO

Limo Provider
mobile base

Replay Mode
--replay

Port C · Transport Layer Transport | WebRTC + AES-128 | SocketCAN · CAN 2.0B · 1 Mbps | 0x151 Motion · 0x155–157 Joint · 0x159 Gripper | serial (Limo)

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L4 · Body Layer

Robot Body

Robots

Robots from different brands can use the same set of capability interfaces. The ones below are the bodies we are currently running.

Computing Board · RDK S100 (D-Robotics) · Ubuntu 22.04 · aarch64

AgileX PiPER In Use
6-DOF arm · gripper
Six-axis arm + gripper, connected to can0.

AgileX NERO In Use
2nd arm
Second robotic arm, connected to can1.

AgileX Limo Pro In Use mobile base
Mobile chassis, serial port driven.

Unitree GO2 Air In Progress
quadruped dog
Quadruped robot dog, WebRTC in progress.

At this moment who controls the body

Control Focus Arbiter

An Android does not have, a robot uniquely possesses a layer. GUI, AI, ring are equal control sources in the Arbiter's eyes; priority **can be declared, can be preempted, can be audited**. When AI is driving the arm, a gesture immediately lets humans override AI, and the arm stops.

SAFETY Safety	3
▲	
HUMAN Human	2
▲	
AI Intelligent Agent	1
▲	
NONE None	0

⟲ Seize Mechanism · MECHANISM

The motion state machine is packed into a **single atomic word** + **revert generation number**; the takeover takes effect immediately at the moment of return.

■ LEVEL-0 · PREEMPT_RT

sched_setattr SCHED_FIFO emergency stop channel, **zero heap allocation**; central collapse or GC, execution layer independent stop, direct to entity, bypass DimOS.

Live Demo · Ring-Driven Robotic Arm

a flick of the hand, the arm follows

Zilo Ring (Move Hand) → nring (Bluetooth Read IMU) → Calculate Pose (RingArmMapper) → Manipulator@1 (MoveToPose) → nervud (Arbitration + Lease) → DimOS (Mechanical Arm Provider) → PiPER (Real Arm Moves)

App only says one sentence "Move your hand here", the rest is left to the system —— **switch to a robotic dog, it's still this sentence**. This is "change the robot, without changing the code".

Nervus OS · ADVX 2026 · repos: **nervud · nervus-ipc · nervus-app-sdk · nervus-packaging · nervus-system-software · nring**

github.com/Nervus-OS · Scan to join the organization ↗